

5.5 Substitution Worksheet

1. Use a u -substitution to compute, if possible, each of the following.

a) $\int_0^2 \frac{7x}{\sqrt{3x^2 + 5}} dx$

b) $\int 3e^{3x} dx$

c) $\int_0^{\pi/3} \tan^3(x) \sec^2(x) dx$

d) $\int 2x \sin(x^2) dx$

e) $\int_0^{10} \frac{\ln(\sqrt{x})}{\sqrt{x}} dx$

f) $\int_0^{\pi/2} 2 \cos(x) e^{\sin(x)} dx$

g) $\int x^2 \sqrt{2+x^3} dx$

h) $\int_e^{e^4} \frac{dx}{x\sqrt{\ln(x)}}$

i) $\int 3x^3 \sqrt{x^2+2} dx$

j) $\int \frac{\ln(x)}{x} dx$

k) $\int_{-\pi/4}^{\pi/4} [x^3 + x^4 \tan(x)] dx$

2. Suppose f is continuous everywhere and $\int_1^4 f(x) dx = 5$. What is the value of $\int_0^{\sqrt{5}} 4x \cdot f(x^2 + 1) dx$?

3. Find the average value of the function $f(x) = \frac{e^{\sqrt{x}}}{\sqrt{x}}$ on the interval $[25, 81]$.

4. Suppose that f is an odd function and g is an even function which are each integrable on the interval $[-1, 1]$. Given that $\int_0^1 f(x) dx = 4$ & $\int_0^1 g(x) dx = 5$, evaluate the following definite integrals.

a) $\int_{-1}^1 [2f(x) + 3g(x)] dx$

b) $\int_{-1}^1 [7 + 3(f(x))^9] dx$

5. Determine the average value of the function $f(x) = xe^{x^2}$ on the interval $[0, 2]$.

6. Use a u -substitution to compute, if possible, each of the following.

a) Evaluate $\int_0^{1/2} \frac{6}{4x^2 + 1} dx$

b) $\int x^3 \tan^2(x^4) dx$.

c) $\int_0^1 x e^{-x^2} dx$

d) $\int_{\frac{1}{2}}^2 \frac{\sec^{-1}(x)}{|x|\sqrt{x^2-1}} dx$

e) $\int \sqrt{2x+1} dx$

f) $\int_1^2 t^2 (2t-1)^6 dt$

g) $\int_{-2}^0 \frac{42x}{(x^2+3)^2} dx$

h) $\int_{-\pi}^{\pi} x \sqrt{x^2 + \cos(x)} dx$